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Full Stack Web Development Report
On

ECO EYE

Submitted by

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Under the Guidance of
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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING



B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
BENGALURU-560019

Sep 2025 – Jan 2026

B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
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Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the project work entitled ECO EYE carried out by Manisha CS(1BM24CS163), Mantri Ramitha(1BM24CS165), Maithri v (1BM24CS159) who are bonafide students of B. M. S. College of Engineering. It is in partial fulfillment for the award of Bachelor of Engineering in Computer Science and Engineering of the Visveswaraiah Technological University, Belgaum during the year 2025-2026. The project report has been approved as it satisfies the academic requirements in respect of Full Stack Web Development(23CS3AEFWD) work prescribed for the said degree.

Signature of the Guide

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Name of the Examiner

Signature with date

B.M.S. COLLEGE OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



DECALARATION

We, Manisha CS(1BM24CS163).Mantri Ramitha (1BM24CS165) Maithri V (1BM24CS159)students of 3rd Semester, B.E, Department of Computer Science and Engineering, BMS College of Engineering, Bangalore, hereby declare that, this Full Stack Web Development entitled ECO EYE has been carried out by us under the guidance of SUPREETH S, Assistant Professor, Department of CSE, BMS College of Engineering, Bangalore during the academic semester Sep 2025 - Dec 2026.We also declare that to the best of our knowledge and belief, the development reported here is not from part of any other report by any other students.

Signature:

MANISHA CS (1BM24CS163)
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INTRODUCTION

Environmental pollution and improper waste disposal have become major issues in urban areas. Overflowing garbage not only affects cleanliness but also causes health hazards and environmental damage. There is a strong need for a digital solution that allows citizens to report garbage issues easily and helps authorities take quick action.

ECO Eye is a full stack web application designed to bridge the gap between citizens and municipal authorities. The system allows users to upload images of garbage found at any location, which can then be accessed by the BBMP (municipal) collector for timely cleanup. Along with this, the platform promotes eco-friendly living by providing an online store for eco-friendly products and spreading environmental awareness.

1.1 Overview

ECO Eye is a full stack web application developed to address the problem of improper waste disposal in urban areas. The system allows users to capture and upload images of garbage found at different locations through the website. These reports can be viewed by BBMP (municipal) collectors, enabling them to take quick action for garbage collection. In addition to this, the application provides information about environmental protection and offers eco-friendly products for purchase through an online cart system. The platform also includes feedback and contact modules to improve user interaction and awareness.

ECO Eye is a web-based application that provides:

- Garbage reporting using image upload
- Location-based issue tracking
- Municipal authority (BBMP collector) access
- Eco-friendly product purchase with cart system
- Environmental awareness information with images
- Feedback and contact modules

1.2 Motivation

Rapid urbanization and population growth have led to a significant increase in waste generation, especially in cities. Improper disposal of garbage in public places, roadsides, and residential areas has become a common problem. This not only affects the cleanliness of the surroundings but also leads to serious health issues such as the spread of diseases, foul odors, and environmental pollution. In many cases, garbage remains unattended for long periods due to lack of proper communication between citizens and municipal authorities.

Although municipal corporations are responsible for waste management, identifying exact locations of garbage and responding in a timely manner remains a challenge. Citizens often do not have an easy or effective way to report garbage-related issues. Traditional complaint systems are time-consuming and do not always provide clear information about the location and severity of the problem. This gap between the public and authorities creates delays in garbage collection and reduces accountability.

Another important motivation for this project is to promote environmental awareness and sustainable living. Many people are unaware of the harmful effects of pollution and the benefits of eco- friendly alternatives. By providing educational content and access to eco-friendly products, ECO Eye encourages users to adopt environmentally responsible habits.

Overall, the project is motivated by the need for a smart, user-friendly solution that supports cleanliness, environmental protection, and collaboration between citizens and municipal authorities.

SOFTWARE REQUIREMENTS

2.1 Hardware Requirements

- A personal computer or laptop with the following minimum or higher specifications:

o **Processor:** Intel Core i3 / AMD Ryzen 3 / Apple M1 or higher
o **RAM:** 8 GB (16 GB recommended for development and multitasking)
o **Storage:** 250 GB SSD or 500 GB HDD
o **Display:** 1366 × 768 resolution or higher
o **Internet Connection:** Stable broadband connection (2 Mbps or higher)

2.2 Software Requirements

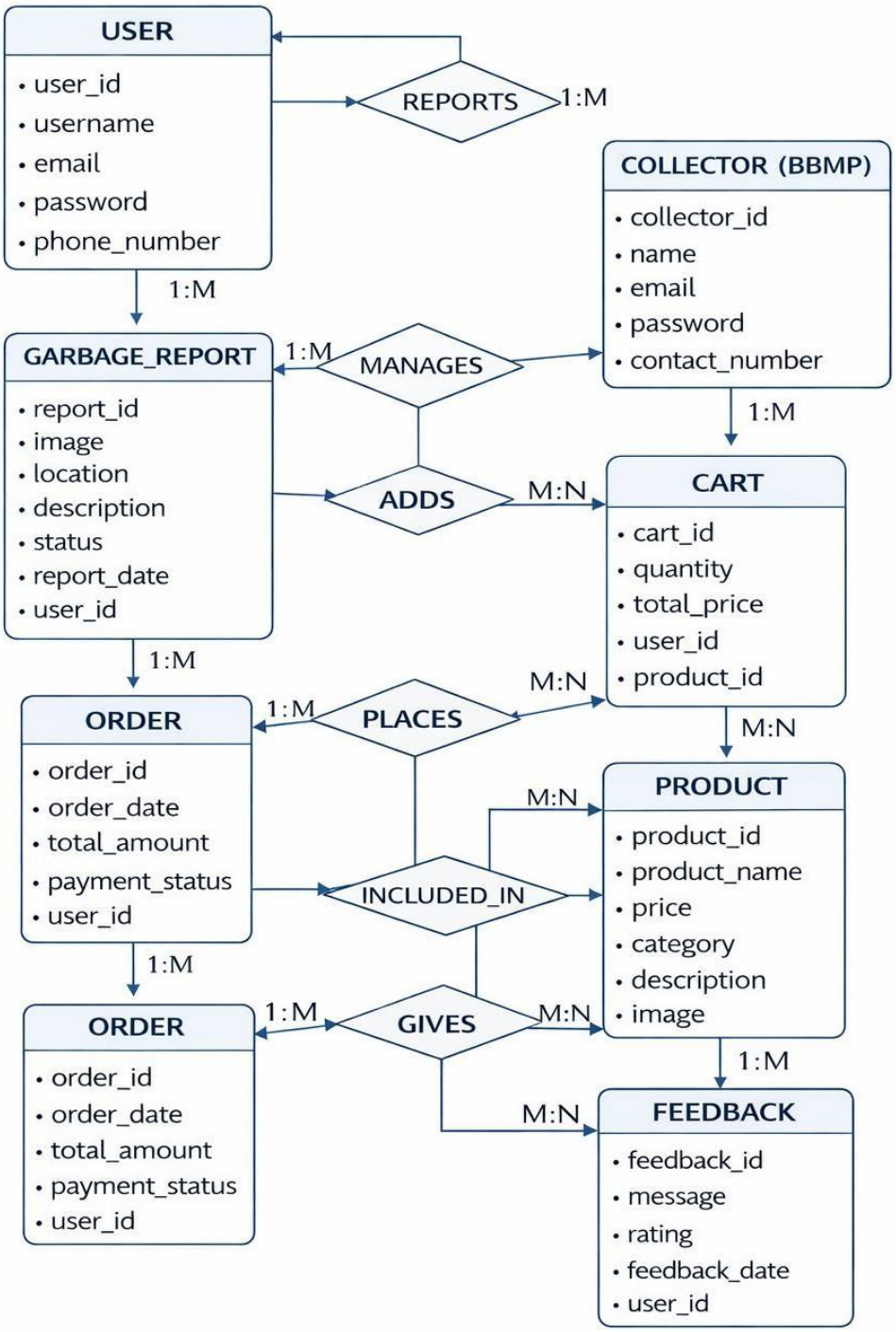
For Windows

- **Operating System:** Windows 10 / Windows 11
- **Front-End Technologies:** HTML, CSS, JavaScript, Bootstrap
- **Back-End Technologies:** PHP or Node.js
- **Database:** MySQL / MongoDB
- **Server:** XAMPP / WAMP / Node.js runtime
- **IDE / Code Editor:** Visual Studio Code, Sublime Text
- **Web Browser:** Google Chrome, Microsoft Edge

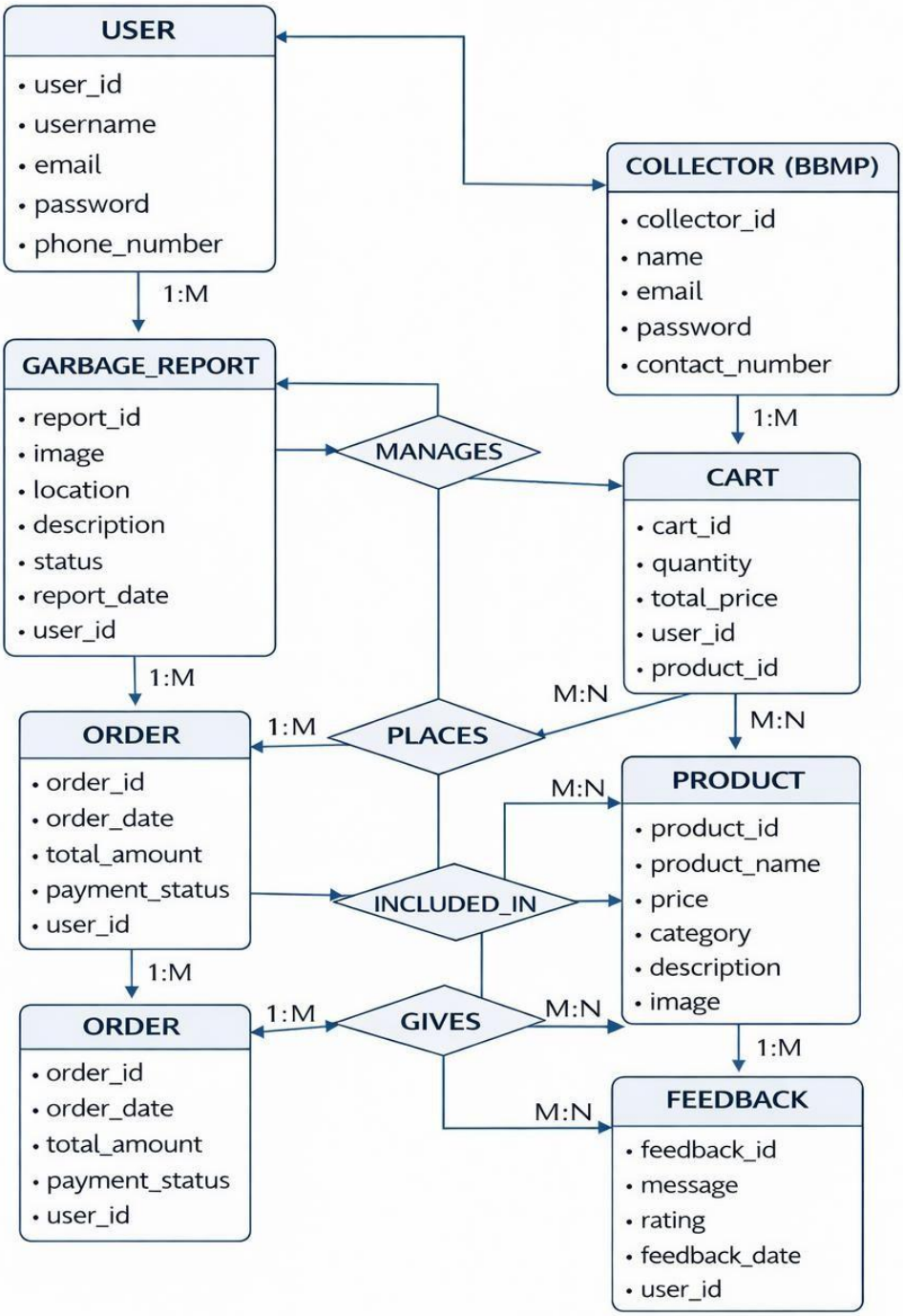
For macOS

- **Operating System:** macOS Monterey or later
- **Front-End Technologies:** HTML, CSS, JavaScript, Bootstrap
- **Back-End Technologies:** PHP or Node.js
- **Database:** MySQL / MongoDB
- **Server:** XAMPP / Node.js runtime
- **IDE / Code Editor:** Visual Studio Code, Sublime Text
- **Web Browser:** Google Chrome, Safari

Entity-Relationship (ER) Diagram of ECO Eye



SCHEMA DIAGRAM OF THE PROJECT



USER INTERFACE DESIGN

Overview

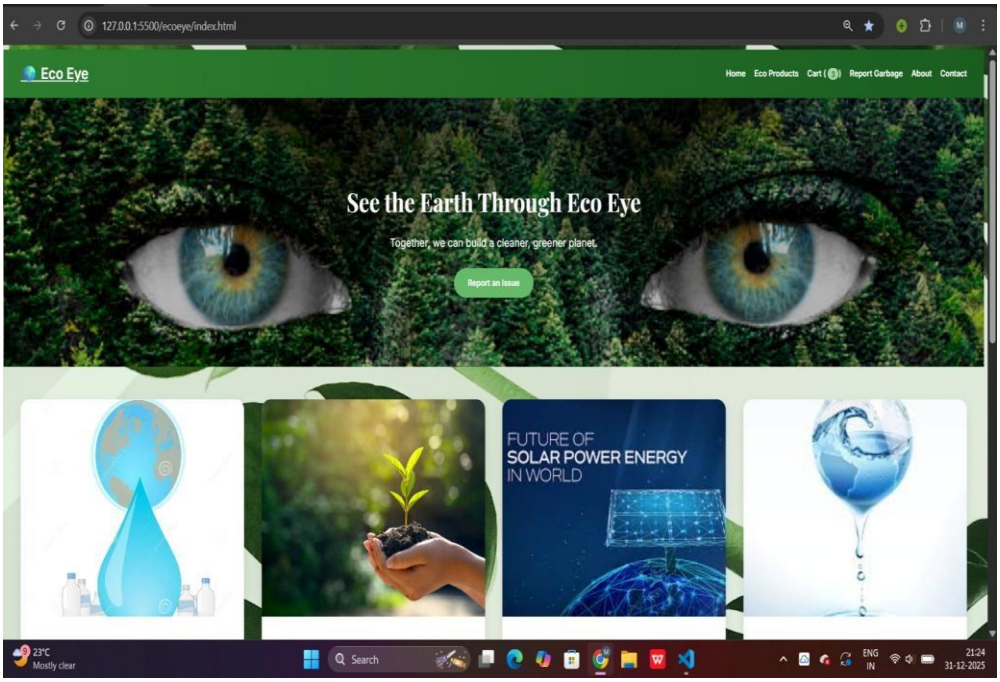


Figure 1: Home Page (ECO EYE Overview)

- The ECO Eye web application features a clean, eco-friendly, and user-centric interface designed to promote environmental responsibility.

- The UI uses green- themed visuals, nature-based backgrounds, simple navigation, and card-based layouts to create an engaging and intuitive experience for users.

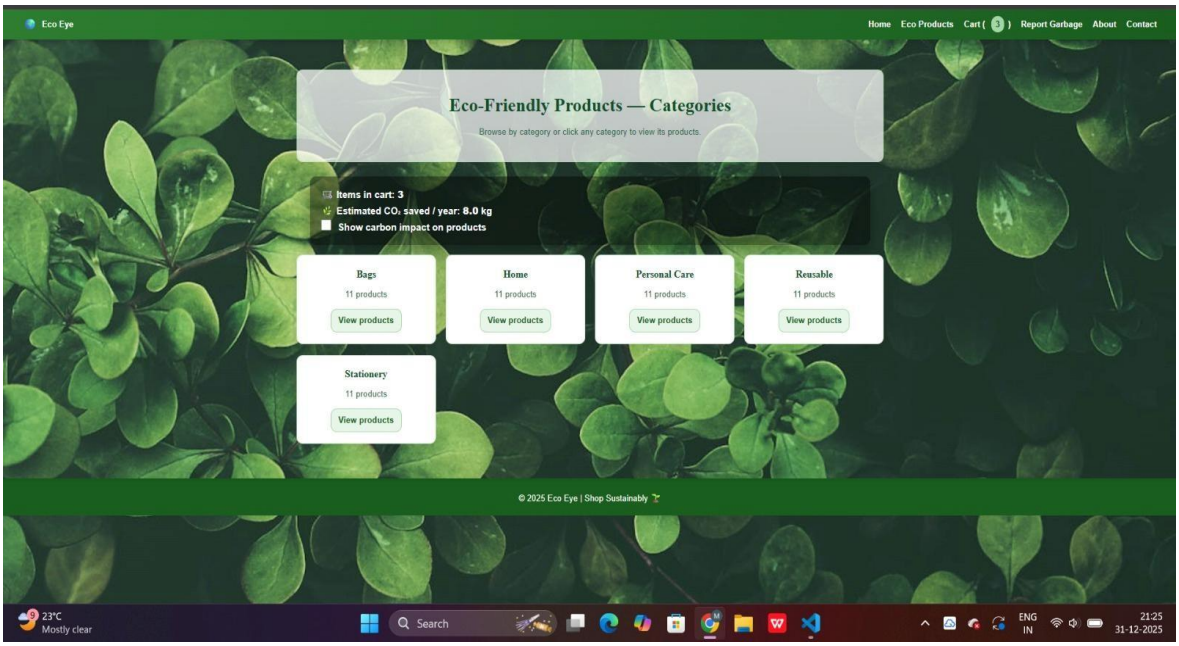


Figure 2: ECO FRIENDLY PRODUCTS CATEGORY

1. Home / Eco Products – Categories Page
Displays eco-friendly product categories such as:

- Bags
- Home
- Personal Care
- Reusable
- Stationery

Each category is shown in a card layout with:

- Category name
- Number of products
- “View Products” button

A carbon impact section shows:

- Items in cart
- Estimated CO_2 saved per year
- Navigation bar includes:
- Home

Eco Products
Cart
Report Garbage
About
Contact

✓ Clean layout improves usability

✓ Carbon-saving info promotes environmental awareness

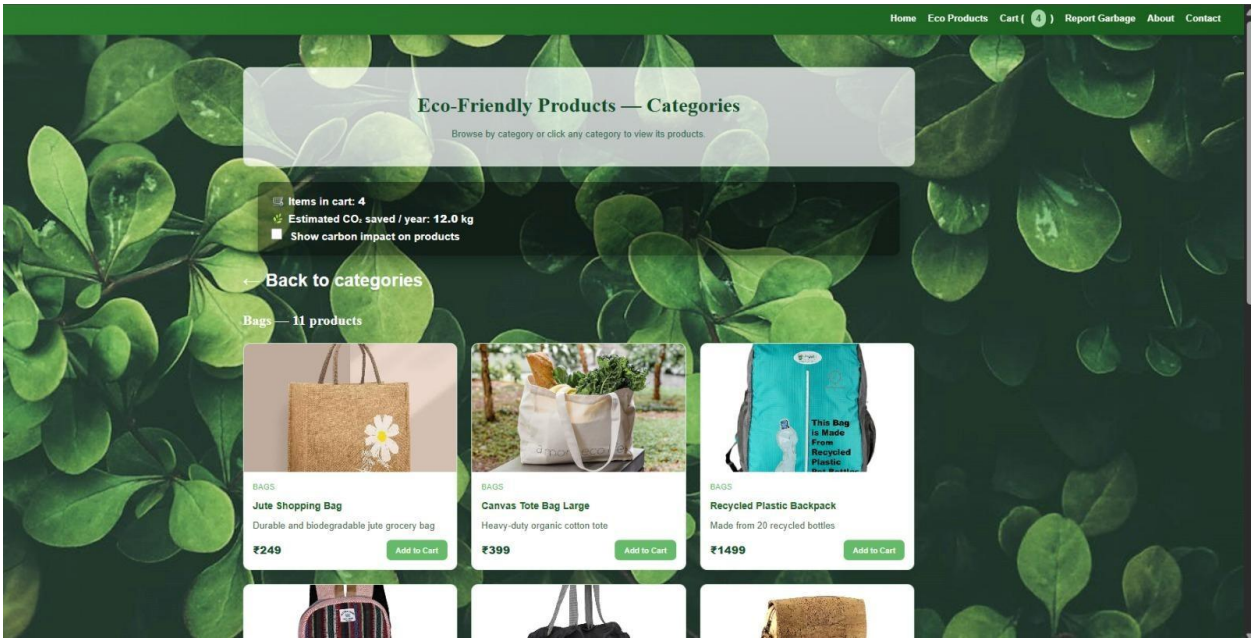


Figure 3: PRODUCT LISTING PAGE

Products are displayed in grid-based cards with:

- Product image
- Product name
- Description
- Price (₹)
- "Add to Cart" button

Users can easily browse and add eco-friendly products to the cart.

"Back to categories" option helps smooth navigation.

✓ Consistent card design

✓ Easy product comparison.

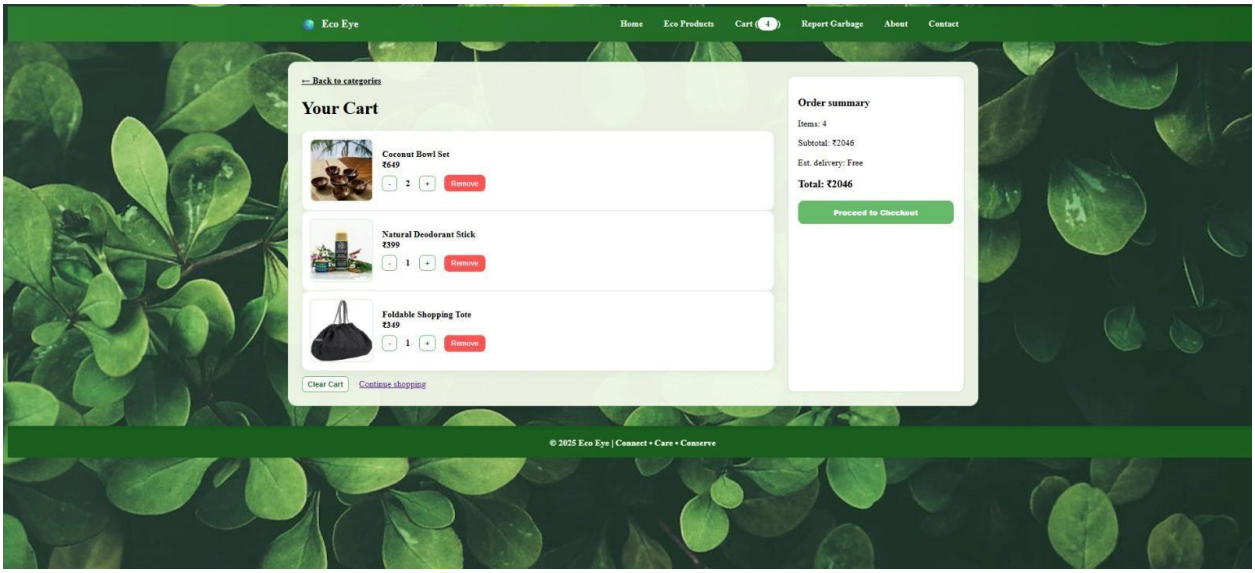


Figure 4: CART PAGE

Shows selected products with:

- Product image
- Quantity controls (+ / −)
- Remove option

Order summary panel includes:

- Total items
- Subtotal
- Delivery status
- Final amount

Buttons:

Proceed to Checkout

Continue Shopping

Clear Cart:

- Simple checkout flow
- Real-time updates for quantity and price

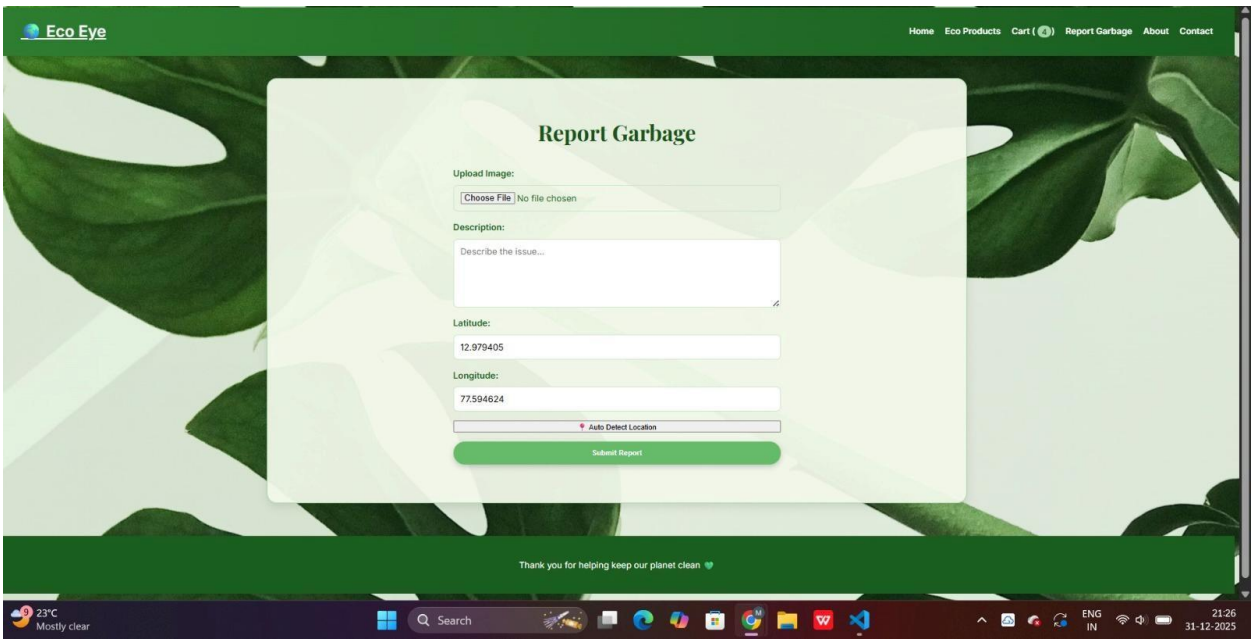


Figure 5: GARBAGE REPORT

Allows users to report garbage at any location.

Features:

- Image upload (garbage photo)
- Description box
- Latitude & Longitude fields
- Auto-detect location button
- Submit Report button

The reported data helps BBMP collectors identify and clean garbage locations.

✓ Encourages community participation

✓ Location-based reporting improves efficiency

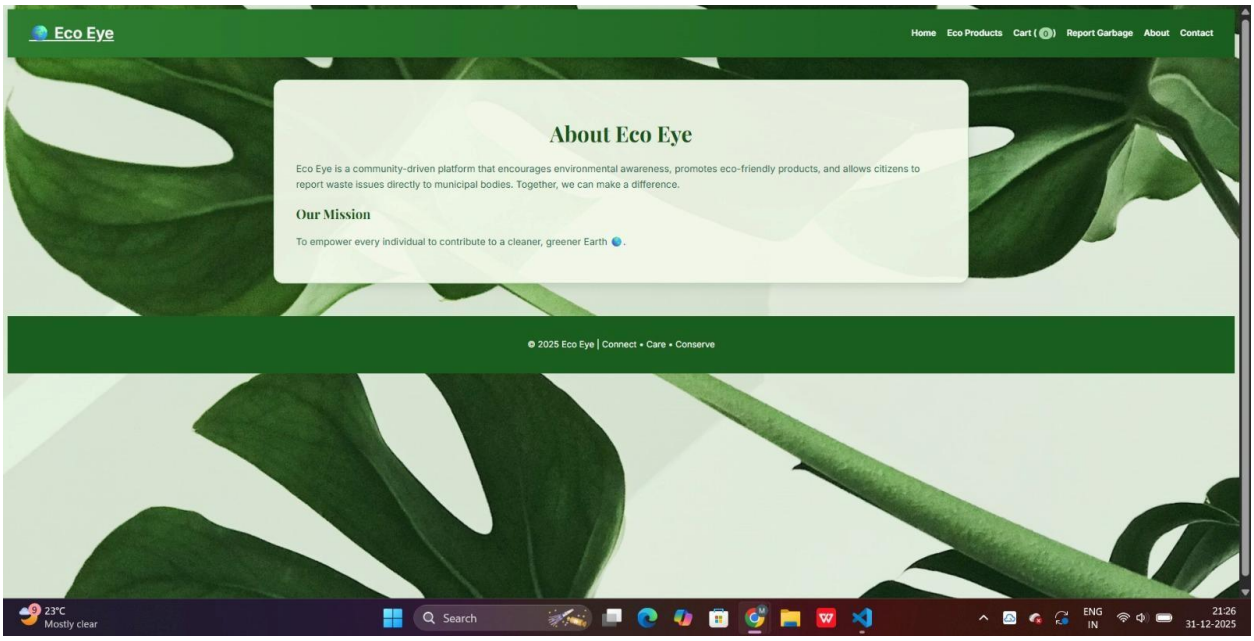


Figure 6: ABOUT ECO EYE

Provides information about ECO Eye:

Purpose of the platform

Community-driven environmental initiative

Mission statement:

“To empower every individual to contribute to a cleaner, greener Earth.”

✓ Minimal design

✓ Clear communication of project vision

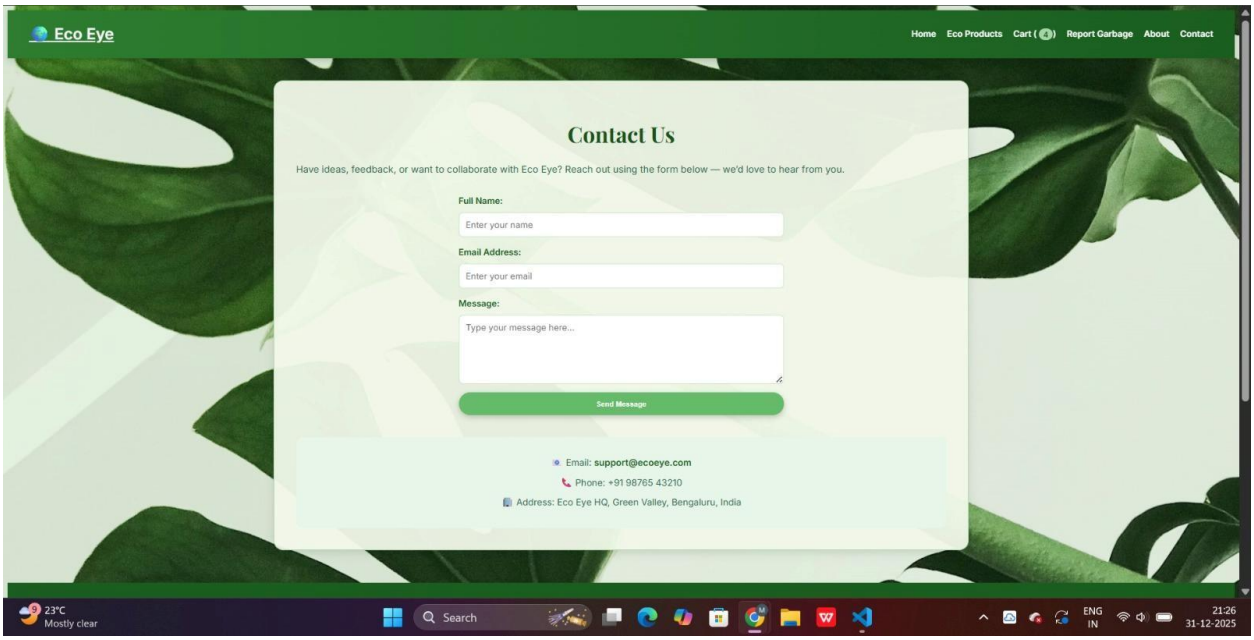


Figure 7: CONTACT US

Contact form includes:
Full Name
Email Address
Message box

Additional details:
Email support
Phone number
Office address

“Send Message” button for user feedback and collaboration.

✓ User-friendly feedback system

✓ Builds trust and transparency

Visual Design & Theme

Green color palette symbolizes sustainability
Nature-inspired background images
Rounded cards and soft shadows for modern look
Responsive layout for different screen sizes

BACK END DESIGN

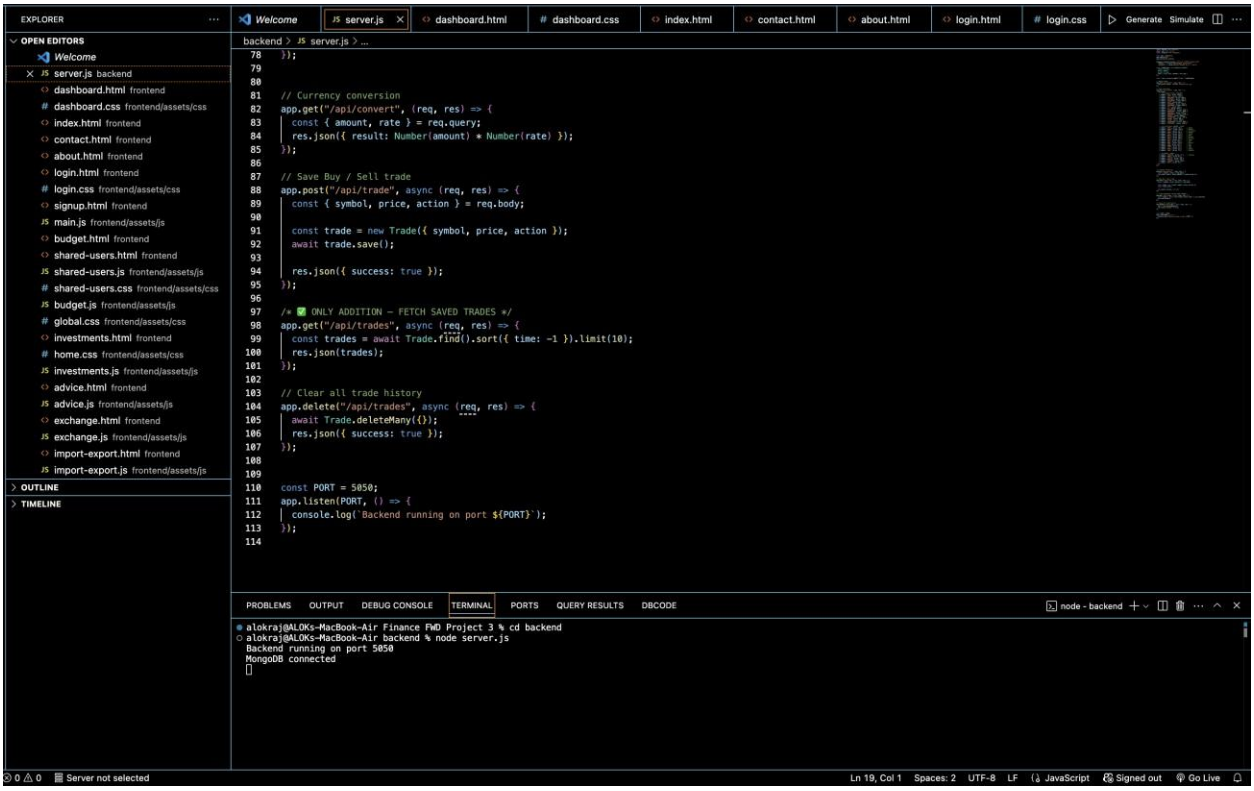


Figure 12: Backend Server Implementation

Overview

The backend of ECO Eye is developed using Node.js, Express.js, and MongoDB Atlas.

It follows a RESTful API architecture to handle user requests such as:

Reporting garbage with location and image

Managing eco-friendly product orders

Handling contact/feedback messages

Connecting frontend with database securely

The backend ensures data storage, API handling, and communication between frontend and database.

Backend Technologies Used

Node.js – Server-side JavaScript runtime

Express.js – Web framework for routing and APIs

MongoDB Atlas – Cloud-based NoSQL database
Mongoose – ODM for MongoDB
CORS – Enables frontend-backend communication
JSON – Data exchange format

API Routes Used

1. Orders API

Route: /api/orders

Purpose:

Store eco-friendly product orders
Manage cart checkout data
✓ Connected to Order.js model
✓ Used during checkout process

2. Garbage Report API

Route: /api/reports
Purpose:
Upload garbage image
Store description and GPS location
Send data to authorities (BBMP)
✓ Supports community-based garbage reporting
✓ Location-based cleanup support

3. Contact / Feedback API

Route: /api/contact
Purpose:
Store user feedback
Handle contact messages
Enable collaboration requests
✓ Improves user interaction
✓ Supports communication with admin

CONCLUSION

The ECO Eye project successfully demonstrates how full stack web development can be used to address real-world environmental challenges. The application integrates garbage reporting, eco-friendly product purchasing, and environmental awareness into a single, user-friendly platform.

Through the Report Garbage feature, users can capture images of garbage along with location details, enabling authorities such as BBMP collectors to identify and clean affected areas efficiently. The eco-friendly products module encourages sustainable living by allowing users to purchase environmentally responsible products through a simple cart and checkout system. Additionally, the Contact and Feedback modules promote user interaction and community involvement.

The backend built using Node.js, Express.js, and MongoDB Atlas ensures secure data handling, scalability, and smooth communication with the frontend. Overall, ECO Eye provides a practical, scalable, and socially impactful solution that promotes environmental cleanliness and sustainability.

FUTURE WORK

The ECO Eye platform can be further enhanced with the following features:

- User Authentication & Authorization
- Implement login and registration using JWT or OAuth
- Separate dashboards for users, administrators, and municipal authorities Real-Time Garbage Tracking System
- Live status updates (Reported → In Progress → Cleaned)
- Notifications to users when garbage is cleared
- Mobile Application Development
- Android/iOS app for easier image upload and location tracking
- Offline reporting support
- AI-Based Garbage Detection-Use image processing and machine learning to automatically classify garbage type
- Priority tagging for hazardous waste
- Admin Dashboard for Authorities
- Visual maps showing garbage hotspots
- Data analytics for cleanup planning
- Online Payment Gateway
- Integrate secure payment options for eco-product purchases
- Order tracking and invoice generation
- Environmental Awareness Section
- Add blogs, tips, and statistics on sustainability
- Carbon footprint tracking per user

REFERENCES

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7)Geolocation API – MDN Web Docs

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8)Environmental Sustainability Concepts

https://www.unep.org